

Definition and first terms

$(5-1\,x)\,A(x)=4+A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*x*T+6*T^2+4*T^3+1*T^4$.
 $a(0),\ldots,a(7)=1,1,7,95,1614,30718,626434,13383650$.
 Kernel used throughout: $\rho(u)=u\frac{1-6\,u^1-4\,u^2-1\,u^3}{1+1\,u}$, so $x=\rho(T(x))$.

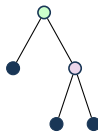
Typogeometry



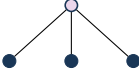
1x1



{1,1}x7



{1,{1,1}}x49



{1,1,1}x4

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 6, Δ_2^{marked} : 1, Δ_3 : 4, Δ_4 : 1.

Coefficient and contour extraction

$$\mathcal{K}_{n,h}=\{(k_1,\ldots,k_3)\in\mathbb{Z}_{\geq 0}^3:h+\sum_{j=1}^3jk_j=n-1\},\quad K=\sum_{j=1}^3k_j,$$

$$a(n)=\frac{1}{n}\sum_{h=0}^{n-1}\binom{n}{h}\sum_{\mathbf{k}\in\mathcal{K}_{n,h}}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_3}6^{k_1}4^{k_2}1^{k_3}.$$

$$a(n)=\frac{1}{2\pi i n}\oint_{\gamma}\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^4Pr(n)a(n+r)=0,\,n\geq 0,\text{ with}$$

$$\begin{aligned}P_0(n)&=3\,(n+1)\,(3\,n-1)\,(3\,n+7)\\P_1(n)&=-15\,(2\,n+3)\,(18\,n^2+54\,n+29)\\P_2(n)&=75\,(n+2)\,(54\,n^2+216\,n+209)\\P_3(n)&=-6750\,(n+2)\,(n+3)\,(2\,n+5)\\P_4(n)&=491\,(n+2)\,(n+3)\,(n+4)\end{aligned}$$

Differential equation

The scalar linear ODE has order 3. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/attached_order4_certificate.json#/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$C(n,t)=\frac{(1+t)P(n,t)}{n(n+1)t^3Q(t)^3}.$$

Telescoping identity: $\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.